Vendor Hotfix and Rollback Ticket: 2025-12-08 Hotfix Summary

Category: Reference | Model: VFIP-CT-HOTFIX-1208
Description: Vendor-issued hotfix documentation and rollback ticket for the post-deployment patch applied on 2025-12-08. Includes hotfix version details, patch notes, list of corrected protective-trip logic, validated test cases, plant application schedule, and post-patch validation checklist.

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1. Executive Summary

On 2025-12-08 the vendor released a hotfix bundle (VFIP XG-440 firmware v3.15.0) to address a regression discovered following rollout of v3.14.2–v3.14.8 during 2025-11-20 → 2025-11-21. The root problem caused intermittent false protective trips (notably TRIP-31 and TRIP-44) under specific reactive-current and frequency step conditions, which resulted in partial plant curtailments and availability degradation. The hotfix restored protective-trip logic and updated internal parameters for ride-through and anti-chatter filtering.

> Scope: Applies to XG-440 inverters across identified serial ranges (see Appendices). Targeted rollout applied to production plants across Zones A/B and a targeted hotfix deployment on 2025-12-08. Vendor acceptance demonstrates partial recovery of availability metrics (per acceptance criteria in Appendix C).

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2. Vendor Release Notes (v3.14.2 → v3.15.0)

The following subsections include precise parameter-level changes and protective-trip behavior modifications contained within the vendor firmware revisions.

2.1 Parameter Changes by Version

Version	Change Type	Parameters / Fields	Notes
v3.14.2 (baseline)	Released	IAntiChatter=120ms; FreqSlopeDeadband=0.15Hz/s	Initial widespread rollout during 2025-11-20 → 11-21. Observed false trips under rapid reactive current changes.
v3.14.9 (interim internal)	Hotpatch (internal)	IAntiChatter adjusted to 80ms for testing	Limited A/B tests—produced unstable behavior in limited edge cases; not released fleet-wide.
v3.15.0	Vendor hotfix (2025-12-08)	IAntiChatter=200ms; FreqSlopeDeadband=0.25Hz/s; TripHoldOff=6000ms (TRIP-31/44 specific); ReactiveRampLimit=15%/s; DiagnosticLogLevel=VERBOSE (conditional)	Addresses anti-chatter, rate-of-change filtering, and extended hold-off for protective trips. Enables conditional verbose logging for 24–72 hrs post-deployment for impacted inverters.

Field Names & Definitions

Field	Datatype	Definition
IAntiChatter	int (ms)	Minimum stable time for current/protection change to avoid chatter
FreqSlopeDeadband	float (Hz/s)	Deadband on measured frequency slope to ignore transient frequency steps
TripHoldOff	int (ms)	Post-detection hold-off preventing immediate re-trip for specified trip codes
ReactiveRampLimit	%/s	Maximum allowed reactive current ramp rate before trip logic engages
DiagnosticLogLevel	enum	OFF/INFO/VERBOSE => controls the amount of logging sent to vendor telemetry

2.2 Corrected Protective-Trip Logic

Below is a concise catalog of protective trip codes, symptoms, root-cause corrections, and new behavior post v3.15.0.

Trip Code	Symptom	Root Cause (pre-fix)	Fix implemented	Status

TRIP-31	Instantaneous inverter shutdown during rapid reactive power commands	Sensitivity to ReactiveRamp > threshold with insufficient anti-chatter window	TripHoldOff extended to 6000ms; ReactiveRampLimit added; IAntiChatter increased	Corrected
TRIP-44	Repeated re-trips after frequency step events causing cascade trips	FreqSlopeDeadband too narrow, causing false detection on legitimate grid events	FreqSlopeDeadband increased to 0.25Hz/s; transient filter applied for 300ms	Corrected
TRIP-12	DC overvoltage false positives observed on high-Irradiance ramps	Telemetry sampling aliasing	Sampling window corrected; added digital low-pass filter to DC sensing	Mitigated
TRIP-70	Grid anti-islanding incorrectly triggered under backfeed test conditions	Threshold arithmetic error in power-imbalance calculation	Fixed arithmetic rounding; added telemetry marker for verification	Corrected

Example Protection Pseudocode (post-fix)

```
// Simplified protective-trip decision-flow (post v3.15.0)
if (time_since_last_trip < TripHoldOff_for_code(code)) {
    suppress_trip(code);
} else {
    if (abs(reactive_ramp_percent_per_s) > ReactiveRampLimit) {
        if (stability_timer >= IAntiChatter) {
            trip(code); // TRIP-31 pathway
        } else {
            log("anti-chatter suppress", code);
        }
    } else if (freq_slope > FreqSlopeDeadband) {
        if (transient_window(<= 300ms)) {
            ignore_event();
        } else {
            trip(code); // TRIP-44 pathway
        }
    }
}
```

2.3 Recommended Rollback Conditions

Vendor-specified automatic rollback triggers for a staged rollout or immediate rollback list.

Condition ID	Trigger	Action	Severity

RB-001	Increase in TRIP-31 events > 5x baseline across a plant within 60 minutes	HOLD further rollouts and initiate immediate rollback to previous stable version (v3.14.2)	Critical
RB-002	Availability drop > 3% absolute for rolling 24-hr window after patch	Pause rollout and run targeted diagnostics; consider rollback per incident review	High
RB-003	Unexpected communication loss to vendor telemetry in > 10% of updated inverters	Isolate update and rollback communications firmware; escalate to vendor NOC	Medium

> Rollback Procedure (concise): - Verify condition matches rollback triggers. - Escalate to vendor support with timestamped logs. - Initiate rollback via vendor portal: set target firmware to v3.14.2, set DiagnosticLogLevel=INFO, disable VERBOSE. - Verify inverters reach RUN state and revalidate protective behavior and availability metrics over 24 hours.

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3. Change Ticket Summaries (2025-11-20 → 2025-12-08)

Summaries pulled from vendor portal: change tickets for initial rollout (2025-11-20 to 2025-11-21) and hotfix (2025-12-08). Each ticket includes scope by plant and inverter serial ranges.

Ticket ID	Date	Summary	Scope (Plants)	Serial Ranges
VT-20251120-01	2025-11-20	Release v3.14.2 staged rollout to Zones A/B	Plant-A1, Plant-A2, Plant-B3	XG440-000100 → XG440-000350
VT-20251121-05	2025-11-21	Completion of v3.14.2 rollout; monitor for protective events	Plant-C1 (pilot)	XG440-000351 → XG440-000380
VT-20251208-HOTFIX	2025-12-08	Deploy v3.15.0 hotfix to affected ranges; enable conditional VERBOSE logs for 72 hrs	Plant-A1, Plant-B3, Plant-D2 (limited)	XG440-000120 → XG440-000220; XG440-000500 → XG440-000560

Change Window & Approval

Hotfix approved by NOC Change Board at 2025-12-07 17:30 UTC. Scheduled application window: 2025-12-08 00:00–06:00 UTC for Plant-A1 and Plant-B3; 2025-12-08 04:00–08:00 UTC for Plant-D2 (low-demand window). Rollback hold window until 2025-12-09 06:00 UTC.

Example Vendor Portal Change Ticket Extract (JSON)

```
{
  "ticket_id": "VT-20251208-HOTFIX",
  "created": "2025-12-07T17:02:00Z",
```

```
  "owner": "vendor.release.team",
  "scope": [
    {"plant": "Plant-A1", "serial_range": ["XG440-000120","XG440-000220"]},
    {"plant": "Plant-B3", "serial_range": ["XG440-000500","XG440-000560"]}
  ],
  "rollback_policy": ["RB-001","RB-002"],
  "log_level": "VERBOSE:72h"
}
```

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4. Deployment & Time-Stamped Logs

The following logs are extracted from the vendor deployment orchestration system and inverters' local logs. Timestamps are UTC.

4.1 Vendor Orchestration Events (Selected)

```
2025-12-08T00:00:02Z - ORCH - VT-20251208-HOTFIX - starting rollout batch 1 - Pla
2025-12-08T00:03:34Z - ORCH - device XG440-000121 - download v3.15.0 - SUCCESS
2025-12-08T00:03:39Z - INVERTER - XG440-000121 - reboot initiated
2025-12-08T00:04:12Z - INVERTER - XG440-000121 - status=RUN, firmware=v3.15.0
2025-12-08T00:14:50Z - ORCH - 5/100 devices reported TRIP-31 within 5 minutes - a
2025-12-08T00:15:01Z - ORCH - check RB-001: TRIP-31 rate=4x baseline - continue r
2025-12-08T02:30:00Z - ORCH - batch1 complete - 95% success (5 devices retried)
```

4.2 Sample Inverter Local Log (XG440-000121)

```
-- 2025-12-08T00:03:39.112Z --
Bootloader v1.4
Applying firmware v3.15.0
-- 2025-12-08T00:04:08.617Z --
Diagnostic: VERBOSE enabled (duration 72h)
Protective: TRIP-31 check => reactive_ramp=18%/s, limit=15%/s
IAntiChatter=200ms satisfied => trip issued (TRIP-31)
-- 2025-12-08T00:05:12.987Z --
TripHoldOff active for TRIP-31: holdoff timer started (6000ms)
-- 2025-12-08T00:05:18.003Z --
Clear: protective reset; state=RUN
```

> Interpretation: The logs show TRIP-31 still triggered on reactive ramp >15%/s, but the TripHoldOff logic prevented immediate re-trip, allowing inverter to return to RUN. This behavior aligns with the design of v3.15.0 to suppress re-trip chatter while still tripping on legitimate threshold breaches.

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5. O&M Incident Report & Postmortem

Consolidated day-by-day timeline, curtailment windows, and action items from operations.

5.1 Timeline (Day-by-Day)

Date (UTC)	Event	Details / Impact

2025-11-20	v3.14.2 initial rollout	Planned rollout; baseline metrics captured. No immediate widespread issues.
2025-11-21	Multiple post-rollout TRIP-31 occurrences	Intermittent curtailment at Plant-A1 & Plant-B3; dispatch curtailments of 2–6 minutes; availability drop ~0.8% per plant.
2025-11-22 to 2025-11-30	Investigation & vendor engagement	Vendor reproduced scenario in lab using rapid reactive injections; confirmed sensitivity in anti-chatter logic.
2025-12-07	Hotfix approved	Change Board approved v3.15.0 deployment window (see Section 3).
2025-12-08	Deployment of v3.15.0	Hotfix deployed; telemetry showed partial reduction in re-trip chatter; some single-event trips persisted with longer hold-offs.
2025-12-09 to 2025-12-11	Post-deployment validation	Availability partially recovered; vendor ongoing work on permanent mitigation (hardware-tolerant sensing patch scheduled Q1 2026).

5.2 Curtailment Window Descriptions

Examples of dispatch curtailment windows observed during incident:

- 2025-11-21 14:12–14:18 UTC — short cascade of TRIP-31 across 8 inverters; automatic curtailment initiated for Plant-A1; total lost energy: 145 kWh.
- 2025-11-22 09:30–09:36 UTC — manual dispatch curtailment to re-synchronize plant with local distribution operator during troubleshooting; no protective trips recorded.

5.3 Root Cause Analysis (RCA)

1. Trigger: Rapid reactive-current command sequences (e.g., AGC/dispatch instructions or local VAR control) produced reactive ramp >15%/s.
2. Failure Mode: Anti-chatter and frequency slope filters in firmware v3.14.x insufficient to suppress detection; protective logic treated legitimate transients as faults.
3. Contributing Factors: High irradiance ramps causing inverter control loops to adjust faster; telemetry aliasing causing sampling misinterpretation for some older hardware revisions.
4. Corrective Action: v3.15.0 increases anti-chatter window, adds reactive ramp limit and extended TripHoldOff, and corrects sampling filter.

5.4 Action Items & Responsible Parties

Action	Owner	Due	Status

Monitor TRIP-31 rates for 72 hours post-hotfix	Operations NOC	2025-12-11	In Progress
Enable VERBOSE logs and collect sample logs for vendor	Vendor NOC / Plant O&M	2025-12-10	Completed
Schedule hardware sampling improvement patch	Vendor R&D	2026-03-15	Planned

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6. Validated Test Cases & Post-Patch Checklist

This section lists the validated test cases executed by vendor and O&M, their pass/fail results, and a practical post-deployment validation checklist for field teams.

6.1 Validated Test Cases (Representative)

Test ID	Description	Procedure	Expected Result	Result
TC-RAMP-01	Reactive Ramp Stress	Stimulate rapid reactive command to inverter: +30% reactive step over 1s (30%/s)	Inverter trips if >ReactiveRampLimit; TripHoldOff prevents chatter; inverter returns to RUN within 10s	PASS (trip occurred, holdoff prevented churn)
TC-FREQ-02	Frequency Step	Apply step change 0.3Hz in <100ms to grid emulator	If freq_slope >0.25Hz/s in 300ms window, transient ignored; no TRIP-44	PASS (no TRIP-44)
TC-DC-03	DC Voltage Sampling	Simulate rapid PV array voltage fluctuation; observe DC sensing and filter	No DC overvoltage false positive; normal power recovery	PASS

6.2 Post-Patch Field Validation Checklist

1. Confirm firmware version: inverter reports `firmware=v3.15.0` in local status
2. Confirm DiagnosticLogLevel is set to VERBOSE for targeted devices (if configured); capture logs for 72h
3. Run TC-RAMP-01 locally on 3 representative inverters per plant during low-demand window; log results
4. Monitor TRIP-31 and TRIP-44 event counts hourly for first 24h; escalate if RB-001 or RB-002 thresholds are met
5. Verify grid telemetry: ensure SCADA shows transient ramps and corresponding inverter telemetry aligned to within 200ms
6. Send confirmation receipts (see Appendix A for sample confirmation format) for all updated serials

6.3 Troubleshooting Flowchart (Textual)

1. Event observed: protective trip (TRIP-31/TRIP-44)
2. Check inverter local log timestamp and error code
3. If reactive_ramp > ReactiveRampLimit, verify if ramp originated from dispatch/AGC — if yes, confirm expected behavior; if no, collect logs and escalate
4. If frequency slope event, check grid disturbance records and apply transient window logic
5. If more than 5 devices in plant trip in 60 min, execute RB-001 rollback workflow

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7. Dispatch Runbooks & CMMS Mapping

Concrete mapping between SCADA alarms, suggested operator actions, and CMMS corrective work order templates.

7.1 SCADA Alarm → CMMS Template Mapping

SCADA Alarm	Severity	Operator Action	CMMS Work Order Template
ALM:PROT_TRIP_TRIP31	High	Verify local inverter state, capture logs, notify vendor; if repeating, issue work order for "Protection Verification"	WO-PROT-VERIFY: include fields: serial, time, trip_count, reactive_ramp_pct_s, snapshot_logs
ALM:PROT_TRIP_TRIP44	High	Check recent frequency events; if coincident with grid event, mark as "grid-related"; else escalate	WO-GRID-INVEST: fields: grid_event_id, frequency_profile, inverter_serials_impacted
ALM:COMM_LOSS_VENDOR	Medium	Retry vendor connection; switch to local control mode if required; issue comms work order	WO-COMM-RESET: fields: modem_status, last_heartbeat, vendor_ticket_id

7.2 Example Corrective Work Order Template (WO-PROT-VERIFY)

```
WorkOrder: WO-PROT-VERIFY
Priority: High
Fields:
```

```
  - site_id: Plant-A1
  - inverter_serial: XG440-000121
  - detected_trip_code: TRIP-31
  - first_detection_time: 2025-12-08T00:14:50Z
  - trip_count_24h: 3
  - reactive_ramp_pct_s: 18
  - firmware_version: v3.15.0
  - attached_logs: /logs/inverter/XG440-000121/2025-12-08_v3.15.0.log
Actions:
  1) Collect local log and SCADA snapshot
  2) Execute TC-RAMP-01
  3) If repeating > 3x, escalate to vendor NOC with ticket VT-20251208-HOTFIX
Notes:
  - Use vendor-provided template for log packaging
  - Attach vendor acceptance return receipt once resolved
```

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8. Settlement & Imbalance Charge Guidance (ERCOT)

Guidance on how penalties are calculated in ERCOT following curtailments and how to reconcile relevant fields.

8.1 Overview of Penalty Calculation

In ERCOT, imbalance charges are assessed when generation deviates from schedules or when Forced Outages impact the settlement. For a generation asset, key settlement impacts include:

- Deviation charges based on real-time price differential for the interval(s) impacted
- Energy Not Served (ENS) calculation for curtailments outside normal dispatch instruction
- Potential Performance Penalties where Availability Drop triggers contractual clauses

8.2 Relevant Data Fields to Reconcile

Field	Source	Notes
SCADA_MW	Plant SCADA (5-min or 1-min depending on metering)	Energy produced per settlement interval; reconcile with ESI meter
METER_MW	Revenue-grade meter	Primary source for settlement; discrepancies require exception report (meter vs scada)
IA_Instruction	ERCOT DA/RTOS instruction feed	Dispatch instructions and curtailed MW amount
Outage_Notice	CMMS / O&M logs	Document reason for curtailment for dispute (e.g., vendor bug)
Settlement_Price	ERCOT RTP/RTBM	Used to calculate imbalance charges per interval

8.3 Example Penalty Calculation (Simplified)

```
// Variables per 5-min interval
scheduled_mw = 50.0;
meter_mw = 45.0;
rt_price = 35.50; // $/MWh

deviation_mw = scheduled_mw - meter_mw; // 5 MW
interval_hours = 5/60; // 0.0833333h
imbalance_cost = deviation_mw * interval_hours * rt_price; // $14.79

// Note: additional settlement rules may multiply by penalty factors depending on
```

8.4 Recommended Reconciliation Procedure

1. Collect interval-level SCADA_MW and METER_MW for affected intervals.
2. Identify intervals where deviations exceed X% threshold (e.g., 3%).
3. Cross-reference with Outage_Notice, change ticket timestamps, and vendor logs to classify deviation cause (vendor, grid, or dispatch).
4. Prepare dispute packet for ERCOT including meter evidence, outage notice, and vendor acceptance for times when equipment was at fault.
5. Track financial impact per plant and apply contractual insurance/penalty offsets as applicable.

9. Vendor Permanent Mitigation & SCADA Threshold Recommendations

Vendor notes and recommended SCADA alarm thresholds to monitor after the hotfix to ensure stability. Also includes acceptance criteria demonstrating partial recovery of availability metrics.

9.1 Vendor Notes on Permanent Mitigation

- Hardware-tolerant sensing firmware scheduled for Q1 2026 (target v3.18.x) to address sampling aliasing in legacy inverter revisions.
- Proposed additional mitigations: adaptive ramp-limiter that correlates AGC/distribution commands and reduces controller aggressiveness during fast irradiance ramps.
- Vendor will provide an extended validation kit for field teams to simulate high ramp scenarios safely in the field.

9.2 Recommended SCADA Alarms & Thresholds

Tag	Threshold	Alarm Level	Action
PROC:REACTIVE_RAMP_PCT_S	> 12%/s	Warn	Log event, notify NOC, start targeted monitoring

PROC:REACTIVE_RAMP_PCT_S	> 15%/s	High	Investigate sources, if sustained >15%/s for >10s openvendor ticket
PROC:TRIP_COUNT_1H (TRIP-31)	> 5	High	Trigger RB-001 workflow
PROC:FREQ_SLOPE	> 0.25Hz/s	Warn	Cross-check with grid event feed
COMM:VENDOR_HEARTBEAT	Missing > 5 min	Medium	Retry comms; generate WO-COMM-RESET

9.3 Vendor Acceptance Criteria (Partial Recovery)

Vendor acceptance criteria used to sign-off hotfix VT-20251208-HOTFIX. Partial recovery is defined as:

1. Reduction in TRIP-31 re-trip frequency by at least 70% within 72 hours post-deployment compared to baseline (pre-hotfix event rates).
2. Plant-level availability returns to within 1% absolute of baseline within 7 days.
3. Telemetry and logs confirming TripHoldOff behaviour and ReactiveRampLimit enforcement on at least 50 sampled events.

> Vendor reported: TRIP-31 re-trips reduced 72% across monitored fleet within 48 hours. Availability improved by 0.9% at Plant-A1 and 0.7% at Plant-B3 during first 72 hours — considered partial recovery pending hardware mitigations.

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10. Appendices: Serial Lists, Confirmation Receipts, Acceptance Criteria

Appendix A — Sample Confirmation Receipt (Per Serial)

```
Confirmation Receipt
Plant: Plant-A1
Inverter Serial: XG440-000121
Firmware Applied: v3.15.0
Applied By: vendor.orch (or local technician)
Applied At: 2025-12-08T00:04:12Z
Post-Validation: TC-RAMP-01 PASS (trip logged; holdoff observed)
Notes: Logs uploaded to vendor portal at /vendor/logs/VT-20251208-HOTFIX/XG440-00
Signature: operations.technician@example.com
```

Appendix B — Affected Serial Ranges (Full)

Plant	Serial Range Start	Serial Range End
Plant-A1	XG440-000100	XG440-000220
Plant-B3	XG440-000500	XG440-000560

Plant-D2	XG440-000800	XG440-000830

Appendix C — Acceptance Criteria & Measured Metrics

Metric	Baseline	Post-Hotfix (72h)	Pass?
TRIP-31 re-trip rate	0.12 events/hr per 100 devices	0.034 events/hr per 100 devices	Yes (72% reduction)
Plant Availability	98.65%	99.55%	Yes (partial recovery)
Telemetry Packet Loss	0.4%	0.5%	No (within tolerance)

Appendix D — Vendor Contact & Escalation Matrix

Role	Name	Contact	Escalation Hours
Vendor NOC	vendor.noc@example.com	+1-555-0100	24x7
Vendor Release Lead	release.lead@example.com	+1-555-0101	Business hours
Vendor R&D	rd.team@example.com	+1-555-0102	Business hours

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Prepared by: Vendor Release & Operations Teams. Confidential within operations, vendor, and authorized stakeholders.
Note: This document is intended for use in operational RAG systems and should be ingested as structured reference for automated decision workflows. Preserve timestamps and code fields when indexing.